

Grade 3 Maths: Angles

Measuring with a Protractor

Subject: Mathematics

Grade: 3

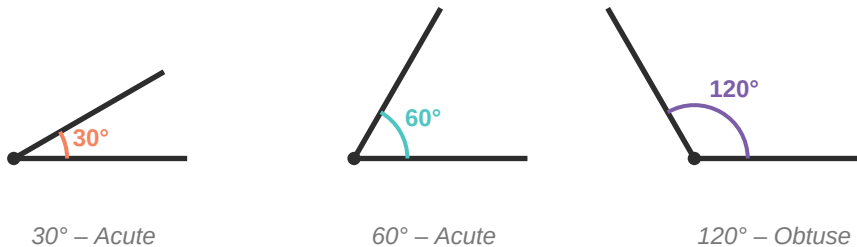
Topic: Angles & Protractors

■ Learning Objectives

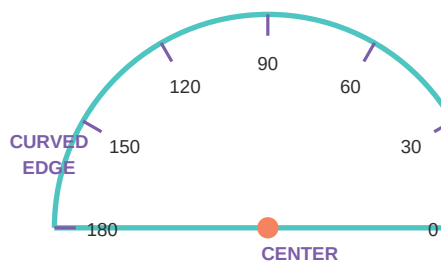
- ✓ Define what an angle is and identify it in everyday objects.
- ✓ Name the parts of a protractor (curved edge, center hole, degree markings).
- ✓ Follow the 4-step method to measure angles accurately.
- ✓ Distinguish between the inner scale and the outer scale.
- ✓ Avoid common measurement mistakes.

■ What Is an Angle?

An **angle** is the space between two straight lines (called **rays**) that meet at a common point called the **vertex**. We measure angles in **degrees (°)**.



■ Meet the Protractor



Center Hole	Place this exactly on the vertex of the angle.
Curved Edge	The semicircular rim that holds the degree markings.
Degree Markings	Numbers from 0° to 180° on both the inner and outer scales.
Baseline	The straight flat edge at the bottom; lined up with one ray.
Inner Scale	Reads left-to-right (0° on the right side).
Outer Scale	Reads right-to-left (0° on the left side).

⇒ ■ How to Measure an Angle – 4 Steps

1

Place the Protractor

Put the **center hole** of the protractor exactly on the **vertex** — the point where the two lines of the angle meet. Make sure it doesn't slip!

2

Align the Baseline

Rotate the protractor so that the **straight edge (baseline)** lines up perfectly with **one ray** of the angle at the **0° line**.

3

Read the Degree Mark

Look where the **other ray** crosses the curved scale. Read the number — that is your angle! Make sure you use the correct scale (inner or outer).

4

Write It Down

Record the measurement and always include the **degree symbol °**. For example: **45°, 90°, 120°**.

■ **Inner vs Outer Scale Tip:** If the angle looks **smaller than 90°**, use the smaller number. If the angle looks **larger than 90°**, use the larger number.

■ ■ Common Mistakes to Avoid

■ Wrong center placement

The center hole must sit exactly on the vertex, not beside it.

■ Baseline not at zero	Always line up one ray with 0° before reading the scale.
■ Reading the wrong scale	Protractors have two sets of numbers. Choose the right one!
■ Protractor slipping	Hold the protractor firmly or use a pencil to mark the vertex first.

■ Tips for Accurate Measuring

- ■ Always double-check the vertex placement before reading.
- ■ Use a sharp pencil to draw clear, thin lines.
- ■ Keep your protractor clean and flat against the paper.
- ■ Practice with many different angles to build confidence.
- ■ Estimate first — does your answer look about right?

■ Fun Facts About Angles!

Angles are all around us every day! The hands of a **clock** form angles, a **pizza slice** is a triangle with angles, **kite tails** make angles in the sky, and even butterfly wings have angles. Can you spot five angles in your classroom right now?

■ Practice Questions

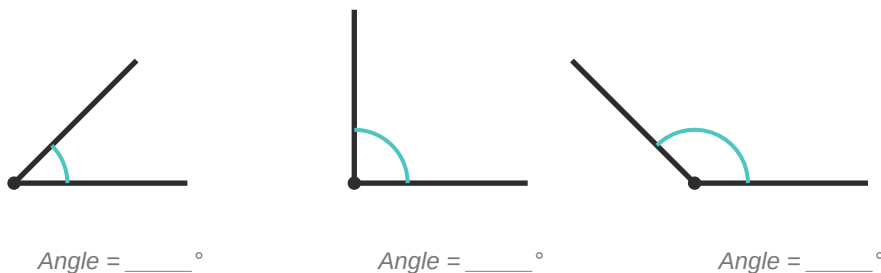
1. What is an angle?

2. Name THREE parts of a protractor.

a) _____ b) _____ c) _____

3. What is the FIRST step when measuring an angle with a protractor?

4. Look at the diagram below. Write the correct measurement for each angle.



5. A student measured an angle and wrote 130° , but the angle looks smaller than 90° . What mistake did they likely make?

6. True or False? You should always write the degree symbol $^\circ$ after your measurement.

Answer: _____

7. Circle the correct answer: An angle that measures exactly 90° is called a a) Straight angle b) Right angle c) Acute angle d) Obtuse angle
8. Draw an angle of 45° in the space below.

■ Answer Key (Teacher Copy)

Q1.	An angle is the space between two lines that meet at a point (the vertex).
Q2.	Any three of: Center hole, Curved edge, Degree markings, Baseline, Inner scale, Outer scale.
Q3.	Place the center hole of the protractor exactly on the vertex of the angle.
Q4.	a) 45° b) 90° c) 135°
Q5.	The student read the wrong scale on the protractor (used the outer scale instead of the inner scale).
Q6.	True
Q7.	b) Right angle

Q8.	Check that the student's drawn angle measures 45° when verified with a protractor.
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■ Summary: 4 Steps to Measure an Angle

- 1■■ Place the center hole on the vertex.
- 2■■ Align one ray with the 0° baseline.
- 3■■ Read the degree mark where the other ray crosses the scale.
- 4■■ Write down the measurement with the $^\circ$ symbol.